QF256: A Fixed-Profile 256-Tone Modem with QRA Coding over GF(256)

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Abstract

QF256 combines a 256-tone, phase-continuous waveform with nonbinary repeat–accumulate coding for 77-bit messages. This article specifies the complete coding and receiver chain and evaluates a fixed implementation on two simulated HF multipath/fading profiles. The study uses 128 frames at each of six noise levels per profile, under unit-average and unit-peak transmit-power constraints. The 1536 randomized cases are reused between the two power settings. Correct-message acceptance is reported by cell with 95% binomial intervals. Under unit-average power, the interpolated 50% acceptance points are approximately -19.4 and -18.8 dB, referenced to 2500 Hz, for the moderate and hard profiles. An independent set of 4096 noise-only 15-second captures yields zero accepted messages and a one-sided 95% upper probability bound of 7.31×10^{-4} per capture. The filtered burst lasts 9.733333 seconds; across the evaluated payload sample its mean 99%-energy bandwidth is 1958.7 Hz and its largest measured peak-to-average power ratio is 0.711 dB. Source, per-trial records, Fortran statistical analysis, and figure-generation code accompany the manuscript. The CRC-conditioned SNR estimator has less than 0.08 dB absolute bias in a fixed-seed AWGN check; dispersive fading produces an uncorrected negative bias that reaches about 1.01 dB in the evaluated hard profile.

1 Introduction

Short-message HF links must recover a small payload despite uncertain arrival time, carrier offset, and time-varying multipath. A noncoherent frequency receiver can supply a full vector of tone likelihoods to a nonbinary code, retaining information that a hard tone decision discards. Factor-graph inference provides the framework for combining those local observations with sparse coding constraints [1]. The QRA construction used here supplies the particular graph and metric machinery [10].

The practical object of study is the complete receiver, including acquisition and message acceptance. A decoder fed a known symbol boundary does not measure this behavior. Likewise, a quoted noise level is ambiguous unless its reference bandwidth, transmit-power constraint, and treatment of fading are specified. QF256 therefore fixes the waveform, coding map, noise units, blind search, and acceptance rule as one reproducible system.

The contribution is a parameter-complete 77-bit implementation and a statistical characterization under two explicitly defined HF channel profiles. The study reports correct acceptance, erasure, erroneous acceptance, noise-only acceptance, and waveform cost. The retained measurements evaluate a fixed receiver; the accompanying analysis regenerates the

results from individual records. Detailed command-line and library contracts are provided in the software documentation [7].

The payload is an opaque sequence of 77 bits whose interpretation is defined by the application. Table 1 lists the fixed parameters.

Quantity	Fixed value
Source payload	77 bits
Tone alphabet	256 frequencies
Symbol duration T_s	0.128 s
Tone separation $\Delta f = 1/T_s$	7.8125 Hz
Complex sample rate f_s	12,000 Hz
Samples per symbol N_s	1536
Transmitted code / pilot symbols	60 / 16
Unfiltered burst duration	$76T_s = 9.728$ s
Transmit FIR	65 taps; 2000 Hz cutoff; Kaiser $\beta = 8.6$
Filtered frame length N_f	116,800 complex samples
Active frame duration $T_f = N_f/f_s$	9.733333333333333 s
Default file capture	180,000 samples; 15 s
Pilot-evidence threshold η_0	22.644583913858970

Table 1: The 12 kHz physical profile.

2 Payload packing and nonbinary coding

2.1 Bit order and CRC

Let $b_1, \dots, b_{77} \in \{0, 1\}$ be the source bits in their public API order. The encoder packs them most significant bit first into ten bytes $m_1, \dots, m_{10}$. The first nine bytes carry 72 source bits. The five remaining source bits occupy the high five bits of m_{10}; its low three bits are fixed to zero. Thus the CRC input is 80 bits: the 77 source bits followed by three public zero bits.

The core computes CRC-16/CCITT-FALSE over these ten bytes, processing each byte from its most significant bit to its least significant bit. The polynomial is `0x1021`, the initial register is `0xFFFF`, and there is no reflection or final XOR. The high and low CRC bytes occupy m_{14} and m_{15}, respectively. The remaining information positions are fixed as

$$m_{11} = m_{12} = m_{13} = 0. \tag{1}$$

2.2 Accumulator construction and transmitted positions

The mother construction has 65 field-valued variables: 15 information positions followed by 50 accumulator positions. Arithmetic uses GF(256) with primitive polynomial `0x11D`; field addition is XOR. Writing the accumulator variables as $a_1, \dots, a_{50}$, the encoder uses

$$a_0 = 0,$$
$$a_i = a_{i-1} \oplus h_i m_{v_i}, \qquad 1 \le i \le 50, \tag{2}$$
$$0 = a_{50} \oplus h_{51} m_{v_{51}}. \tag{3}$$

The fixed graph has 51 check nodes and 151 edges. Its coefficients satisfy the termination identity in (3). Appendix A records the graph indices and coefficient construction used by the source.

The following 60 variables select data tones, in this order:

$$(m_1, \ldots, m_{10}, a_1, \ldots, a_{50}). \tag{4}$$

The three zero information symbols in (1) are *shortened*: the decoder knows their values. The two CRC symbols m_{14}, m_{15} are *punctured*: they are not transmitted, and their initial decoder priors are uniform. Their values must be recovered through the code constraints before the CRC comparison can pass. The low three bits of m_{10} are also constrained to zero in the decoder prior.

2.3 The operational codebook

The mother graph is linear over GF(256): multiplication by a fixed coefficient and XOR preserve field addition. The operational encoder applies additional bit and CRC constraints before selecting the transmitted positions. Let $F : \{0,1\}^{77} \to \mathrm{GF}(256)^{60}$ denote that complete map. The first ten transmitted symbols retain every input bit, so F is injective and its image contains exactly 2^{77} words.

This image is not a generic linear code over GF(256). The low-three-bit restriction is not closed under arbitrary field multiplication, and the CRC with initial register `FFFF` is affine in its input bits. In the polynomial bit representation, fixed field multiplication, packing, and puncturing are binary-linear operations; their composition with this CRC makes F a binary-affine map. Cardinality and injectivity do not specify its minimum distance. The ratio $77/(60 \cdot 8) = 77/480$ counts source bits per transmitted binary-label position, excluding pilots; it is not a field-symbol dimension.

The graph and coefficients are adapted from the attributed QRA source material. The GF(256) construction uses the numeric coefficients specified in Appendix A [6, 10].

3 Frame allocation and transmit waveform

3.1 Public pilots and tone mapping

Number frame symbols from 1 through 76. The sixteen pilots occupy

$$p_j = 1 + 5(j-1), \qquad j = 1, \ldots, 16. \tag{5}$$

The remaining positions carry the 60 values in (4). The layout is fifteen groups of one pilot followed by four data symbols, then one final pilot, as illustrated in Figure 1.

The pilot tone sequence is deterministic and public. Starting with $r_0 = 167$, it is generated by

$$r_j = (109 r_{j-1} + 89) \bmod 256, \qquad j = 1, \ldots, 16. \tag{6}$$

For any pilot or data value $m \in \{0, \ldots, 255\}$, the baseband frequency is

$$f_m = (m - 127.5)\Delta f, \qquad \Delta f = 7.8125\,\mathrm{Hz}. \tag{7}$$

The extreme tone centers are $-996.09375\,\mathrm{Hz}$ and $+996.09375\,\mathrm{Hz}$. Their span is $1992.1875\,\mathrm{Hz}$; the convention $256\Delta f$ gives a nominal 2 kHz tone lattice. Filtering and finite burst duration also determine the actual occupied spectrum.

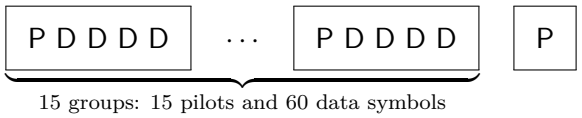


Figure 1: The 76-symbol frame. P is a public pilot and D is a transmitted code symbol. Complete filter tails extend the output beyond the symbol train.

3.2 Phase continuity, filtering, and normalization

Before filtering, each symbol is a unit-magnitude complex sinusoid. Since $f_m T_s = m - 127.5$, every tone advances the phase by an odd multiple of π during one symbol. The boundary phasor therefore alternates between $+1$ and -1 independently of the data. The implementation uses that property to generate a continuous-phase sequence at tone transitions.

The finite symbol train is convolved with a real, symmetric 65-tap low-pass kernel. At the public sample rate, let $q = -32, \ldots, 32$, $f_c = 2000\,\mathrm{Hz}$, and $\beta = 8.6$. Before DC-gain normalization, its samples are

$$\widetilde{h}[q] = \begin{cases} 2f_c/f_s, & q = 0, \\ \sin(2\pi f_c q/f_s)/(\pi q), & q \neq 0 \end{cases} \frac{I_0\left(\beta\sqrt{1-(q/32)^2}\right)}{I_0(\beta)}, \tag{8}$$

with $h[q] = \widetilde{h}[q]/\sum_{u=-32}^{32} \widetilde{h}[u]$. Here I_0 is the modified Bessel function. The causal stored kernel has 32 samples of group delay. Full convolution is retained: the 116,736-sample symbol train gains 64 samples, producing all 116,800 output samples without cropping either tail.

If $z[n]$ is this filtered frame, the transmitter returns

$$x[n] = \frac{z[n]}{\sqrt{N_f^{-1} \sum_{k=0}^{N_f-1} |z[k]|^2}}. \tag{9}$$

Consequently the mean complex power of the complete active frame is one in the core's floating-point representation. Filtering introduces instantaneous envelope variation [5].

4 Receiver processing

The receiver accepts complex IQ at the fixed sample rate, estimates noise from the supplied capture, determines time and frequency offsets, measures tone energies, and decodes one codeword. It returns at most one message per capture. Figure 2 summarizes the sequence.

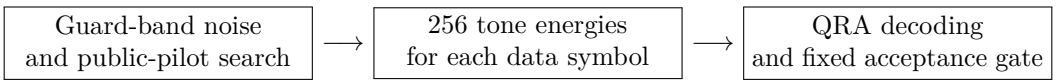


Figure 2: Receiver processing. The caller supplies the IQ capture and chooses known or blind synchronization.

4.1 Noise estimate and units

For the sample-domain convention used here,

$$\mathbb{E}\{|w[n]|^2\} = f_s N_0. \tag{10}$$

The receiver divides the input into complete, nonoverlapping 512-sample blocks. Each block is multiplied by the periodic Hann window $u[n] = \frac{1}{2} - \frac{1}{2}\cos(2\pi n/512)$ and transformed with an unnormalized FFT. Positive and negative bins in the public 2500–5000 Hz guard bands are pooled; at 12 kHz these correspond to signed bin indices $\mathcal{K} = \{\pm 107, \ldots, \pm 213\}$.

If $X_b[k]$ denotes a windowed block spectrum, B is the number of complete blocks, and $W = \sum_{n=0}^{511} u[n]^2$, the estimate is

$$\widehat{N}_0 = \max\left[\frac{\sum_{b=1}^{B}\sum_{k\in\mathcal{K}}|X_b[k]|^2}{B|\mathcal{K}|W f_s},\ 10^{-20}\right]. \tag{11}$$

The public result field `noise_variance` is $f_s\widehat{N}_0$. The estimator assumes that noise power in these guard bands represents the noise in the tone region. Input filtering must retain both guard bands for this estimate; interference or colored noise changes the measured value.

4.2 Time and frequency acquisition

For a candidate zero-based burst offset s and residual frequency ν, the pilot score is

$$S(s,\nu) = \frac{1}{f_s N_s}\sum_{j=1}^{16}\left|\sum_{n=0}^{N_s-1} y[s+32+(p_j-1)N_s+n]\, e^{-\mathrm{j}2\pi(f_{r_j}+\nu)n/f_s}\right|^2. \tag{12}$$

The 32-sample offset aligns the rectangular tone integration with the transmit filter's group delay.

Blind acquisition first searches starts from 0.3 to 0.7 seconds in 96-sample steps and frequencies from -10 to $+10$ Hz in 0.5 Hz steps. It then refines around the winning start by at most 96 samples in either direction, using 4-sample steps, and around the winning frequency by at most 0.5 Hz, using 0.025 Hz steps. The refinement remains clipped to the search bounds. The maximum pilot score selects the acquisition result.

Known synchronization evaluates only $s = 0, \nu = 0$. It is appropriate when the caller has already aligned the burst and removed the residual carrier offset.

4.3 Tone energies and soft metrics

At the selected offset and CFO, the receiver measures the 256 candidate tone energies in each of the 60 data-symbol intervals, with the same $1/(f_s N_s)$ normalization as (12). The implementation uses three 512-point FFT lanes to evaluate the required bins of the 1536-sample transform. Under the convention in (10), a unit-energy tone projection has complex noise variance N_0.

The core uses a fixed noncoherent intrinsic metric. Let $E_{m,i}$ be the energy for tone m in data interval i, and define

$$\gamma = 6\left(\frac{13}{63}\right)10^{0.28}, \qquad z_{m,i} = 2\sqrt{\gamma E_{m,i}/\widehat{N}_0}, \tag{13}$$

$$L(z) = \frac{z^2(z+0.039)}{0.9931z^2 + 2.6936z + 0.5185}. \tag{14}$$

The rational expression approximates $\log I_0(z)$ and is part of the attributed QRA metric implementation. For each interval, $L(z)$ is shifted by its maximum, floored at -700 in the exponent, exponentiated, and normalized over the tone alphabet. These probabilities are the decoder observations [6, 10].

4.4 Iterative decoding and final acceptance

The decoder performs sum–product updates over the fixed nonbinary graph, using Walsh–Hadamard transforms to evaluate additive-field convolutions [1, 6]. It runs at most 100 iterations [6]. The transmitted symbols receive their measured priors; the known zero positions and the three low zero bits of m_{10} are constrained as described in Section 2.

Before declaring a raw successful decode, the implementation requires an extrinsic-convergence score of at least $65 - 0.01$, zero graph syndrome, and agreement of the recomputed CRC bytes with the recovered punctured CRC positions. The CRC is a final check, not an additional factor enforced during message passing. The convergence score is the sum, over all 65 variables, of each normalized maximum extrinsic probability.

The application acceptance test adds a pilot gate:

$$\eta = S(\widehat{s}, \widehat{\nu})/\widehat{N}_0 - 16, \tag{15}$$

$$\text{accepted} \iff \text{crc_ok} \ \wedge \ (\eta \geq \eta_0), \qquad \eta_0 = 22.644583913858970. \tag{16}$$

Here `crc_ok` denotes the complete raw core success condition, including convergence and syndrome checks. Both file frontends use the same threshold η_0.

4.5 CRC-conditioned SNR estimate

For a CRC-valid decode, let $\widehat{t}_i$ be the recovered tone in data interval i and $E_{\widehat{t}_i,i}$ its received matched-projection energy. The receiver regenerates a clean complete frame from that payload, normalized to unit mean complex power by (9), and measures the corresponding correct-bin energies $E^{(1)}_{\widehat{t}_i,i}$. The estimated received active-frame power and reported SNR are

$$\widehat{P} = \frac{\sum_{i=1}^{60} (E_{\widehat{t}_i,i} - \widehat{N}_0)}{\sum_{i=1}^{60} E^{(1)}_{\widehat{t}_i,i}}, \tag{17}$$

$$\widehat{\mathrm{SNR}}_{2500} = 10 \log_{10} \left(\frac{\widehat{P}}{2500 \widehat{N}_0} \right). \tag{18}$$

In aligned static AWGN, a correct-bin energy has expectation $P E^{(1)}_{\widehat{t}_i,i} + N_0$. Subtracting one $\widehat{N}_0$ per interval therefore estimates received signal power without treating the pilot evidence as SNR. The public validity flag is set only when the CRC is valid and the noise, reference sum, signal sum, and final logarithm are finite with positive power. No Doppler or SNDR estimate is reported.

5 Performance-study method

5.1 Channel and observation model

The study uses two Gaussian-scatter tapped-delay profiles in the spirit of stationary HF channel modeling [2]. Their exact implementation is defined here, rather than identified with a standardized profile. Let $x[n]$ be the normalized transmit frame, extended by zero outside its support, and let A apply the chosen transmit-power constraint. With $t = (n - s)/f_s$, the received capture is

$$y[n] = A e^{\mathrm{j}2\pi\nu t} \sum_{\ell=1}^{2} \sqrt{P_\ell}\, g_\ell(t) x[n - s - d_\ell] + w[n], \qquad 0 \leq n < 180000. \tag{19}$$

Here s is an integer start offset, ν is constant CFO, and the path processes are mutually independent. Each has the finite spectral representation

$$g_\ell(t) = \sum_{k=1}^{64} c_{\ell k} e^{\mathrm{j}2\pi F_{\ell k} t}, \quad c_{\ell k} \sim \mathcal{CN}(0, 1/64), \quad F_{\ell k} \sim \mathcal{N}(0, (B_D/2)^2). \tag{20}$$

Averaging over both coefficients and frequencies gives $\mathbb{E}|g_\ell|^2 = 1$ and correlation $R_g(\tau) = \exp(-\pi^2 B_D^2 \tau^2/2)$. A realization contains 64 spectral lines; it approximates the continuous Gaussian spectrum. Individual fades are not power-normalized.

Profile	Delays	Path powers	$B_D = 2\sigma_F$
Moderate	0, 1 ms (0, 12 samples)	1/2, 1/2	1 Hz
Hard	0, 2 ms (0, 24 samples)	1/2, 1/2	4 Hz

Table 2: The two simulated HF profiles at $f_s = 12$ kHz.

The delays in Table 2 are exact integer sample shifts. Fading is evaluated from the frame origin, and delayed tails are retained. The start is drawn uniformly from $[0.4, 0.6)$ s and rounded to the nearest sample; CFO is uniform over $[-2, 2)$ Hz. There is no frequency drift. The receiver receives only the capture and fixed public configuration; it searches the larger region specified in Section 4.

Noise is independently generated for all 180,000 capture samples, including samples outside the burst. For a reference level ρ in dB,

$$N_0 = \frac{1}{2500\, 10^{\rho/10}}, \qquad w[n] \sim \mathcal{CN}(0, f_s N_0). \tag{21}$$

Each real quadrature has variance $f_s N_0/2$. Noise is not rescaled by the instantaneous faded receive power or by the waveform bandwidth.

5.2 Transmit power, energy, and waveform measurements

In the unit-average setting $A = 1$, so the active-frame transmit power $P_{\mathrm{av}} = 1$ and energy $E = \sum |Ax[n]|^2/f_s = T_f$. In the unit-peak setting, $A = 1/\sqrt{P_{\mathrm{pk,ref}}}$, where $P_{\mathrm{pk,ref}}$ is the larger of the peaks obtained by regenerating the waveform at 48 and 192 kHz, normalized by the 12 kHz frame's pre-normalization mean power. The 48 and 192 kHz FIRs contain 257 and 1025 taps, respectively, preserving the kernel support of $\pm 32/12000$ s in (8) [5]. The check requires the two peak estimates to agree within 0.05 dB. Consequently, $P_{\mathrm{av}} = 1/P_{\mathrm{pk,ref}}$ and $E = T_f/P_{\mathrm{pk,ref}}$ in that setting. These are measured oversampled peaks, not a proof of a continuous-time supremum.

The horizontal coordinate ρ always denotes the unit-reference level in (21). Actual average transmit SNR and source-bit energy are

$$\mathrm{SNR}_{2500} = \rho + 10\log_{10} P_{\mathrm{av}}, \qquad (E_b/N_0)_{77} = \rho + 10\log_{10}(2500E/77). \tag{22}$$

The first equals ρ only under unit-average normalization. In the unit-peak setting it is approximately $\rho - 0.711$ dB. These quantities refer to transmit power and ensemble channel normalization, not to the power of a particular faded receive frame. Dividing by 77 counts supplied payload bits; the sampled payload entropy is 71 bits as described below.

Bandwidth is measured on the complete active waveform before the channel. An unwindowed FFT is zero-padded to the next power of two at least $4N_f$ (524,288 points here). The ordered, two-sided squared spectrum defines cumulative-energy quantiles $f_{0.005}$ and

$f_{0.995}$, interpolated within bins. The reported width is $B_{99} = f_{0.995} - f_{0.005}$; FFT spacing is approximately 0.0229 Hz. PAPR is $10\log_{10}(P_{\mathrm{pk,ref}}/P_{\mathrm{av,12k}})$ for the unscaled waveform. Here $P_{\mathrm{av,12k}}$ is the mean power of the normalized 12 kHz frame, approximately one, before unit-peak scaling. Both width and PAPR are invariant under the subsequent constant peak scaling.

5.3 Sampling and fixed receiver configuration

For each profile, the reference levels are $\{-24, -22, -20, -18, -16, -14\}$ dB, with 128 separately seeded frames in every cell. The two power settings reuse the payload, fading, noise, start, and CFO draws: there are 1536 underlying randomized cases and 3072 outcome records, not 3072 independent realizations. Outcomes within a cell use different payload, channel, and noise seeds. The analysis makes no independence assumption between power settings.

The message generator draws 71 random bits and appends the fixed suffix `101000`. It consumes 18 uniform draws to form the random part; the next two draws set start and CFO. Thus the error-rate experiment samples a defined subset of the encoder's 2^{77} codebook. With profile index $q \in \{3, 4\}$, SNR index $j \in \{1, \ldots, 6\}$, trial $i \in \{1, \ldots, 128\}$, $c = 6(q-1)+j$, and $b = 20261201$, the seeds are

$$\begin{aligned} s_p &= b + 100003c + 101i, \\ s_h &= b + 200003c + 307i, \\ s_w &= b + 300007c + 911i + 29. \end{aligned} \tag{23}$$

The Fortran random generator is initialized separately for these streams; Gaussian values use the documented Box–Muller transform. Bitwise replay therefore depends on the retained GNU Fortran runtime convention. The standalone harness implements the same initialization and draw order.

The waveform, QRA graph, metric, iteration limit, acquisition bounds, sample counts, channel settings, and confirmation grid were fixed before the reported confirmation outcomes were inspected. Only the pilot-evidence threshold was calibrated, using a separate 2048-capture noise lot. Its type-7 empirical 99th percentile is

$$\eta_0 = 0.47\eta_{(2027)} + 0.53\eta_{(2028)} = 22.644583913858970, \tag{24}$$

where parentheses index the sorted calibration evidence values. The threshold and records were hashed before the independent noise check and confirmation analysis. It was not retuned using the reported outcomes [8].

5.4 Endpoints and uncertainty

A signal trial succeeds only when the final acceptance rule passes and all 77 recovered bits equal the transmitted bits. An erasure is a trial with no accepted message; an incorrect acceptance passes the rule with different bits. These three counts sum to the cell size. Frame error rate is one minus correct acceptance, so it includes both erasures and incorrect acceptances.

For k correct acceptances among $n = 128$ trials, $\widehat{p} = k/n$. The pointwise two-sided 95% Wilson interval [3] is

$$\frac{\widehat{p} + z^2/(2n) \;\pm\; z\sqrt{\widehat{p}(1-\widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}, \qquad z = 1.95996398454. \tag{25}$$

These are cell intervals, not simultaneous confidence bands. The grid average weights all twelve profile/SNR cells equally and is reported as a descriptive summary, without treating heterogeneous cells as one binomial population.

The estimated 50% crossing linearly interpolates the two adjacent empirical rates that bracket 0.5. It is reported to 0.1 dB because the grid spacing is 2 dB. To express sampling uncertainty without assuming a subgrid curve shape, each curve also receives a conservative grid bracket. Exact binomial intervals use a Bonferroni allocation across its six points (tail probability 0.05/12). The bracket extends from the highest grid point whose upper bound is below 0.5 to the lowest whose lower bound exceeds 0.5. Its confidence interpretation requires monotone success probability in SNR; it is not an interval for the linear-interpolation model.

The independent noise-only check uses 4096 fresh 15-second captures and retains the same blind receiver. Reference levels cycle through the six noise scales in (21); there is no received SNR without a signal. Calibration and check seeds are disjoint. For zero accepted messages, the one-sided exact 95% binomial upper bound is [4]

$$p_U = 1 - 0.05^{1/4096} = 0.0007311125565. \tag{26}$$

It is exact under a common capture probability. If probabilities differ between noise scales, it is conservative for their allocation-weighted mean: for independent captures, $\prod_i (1 - p_i) \leq (1 - \overline{p})^{4096}$. Calibration samples are excluded from this denominator. The supplied provenance also records limited scalar-seed coincidences with signal-payload seeds; the study does not assume complete mutual stream independence across all lots.

6 Results

6.1 Correct-message acceptance and erasures

Figure 3 and Table 3 report every confirmation cell. Under unit-average power, the equal-weight grid summary is 784 correct acceptances in 1536 records (51.0%); under unit-peak power, it is 686/1536 (44.7%). The profile totals are 414/768 and 370/768 for moderate and hard fading at unit-average power, and 367/768 and 319/768 at unit-peak power. These fractions summarize this deliberately balanced SNR grid, rather than an operating distribution of HF conditions.

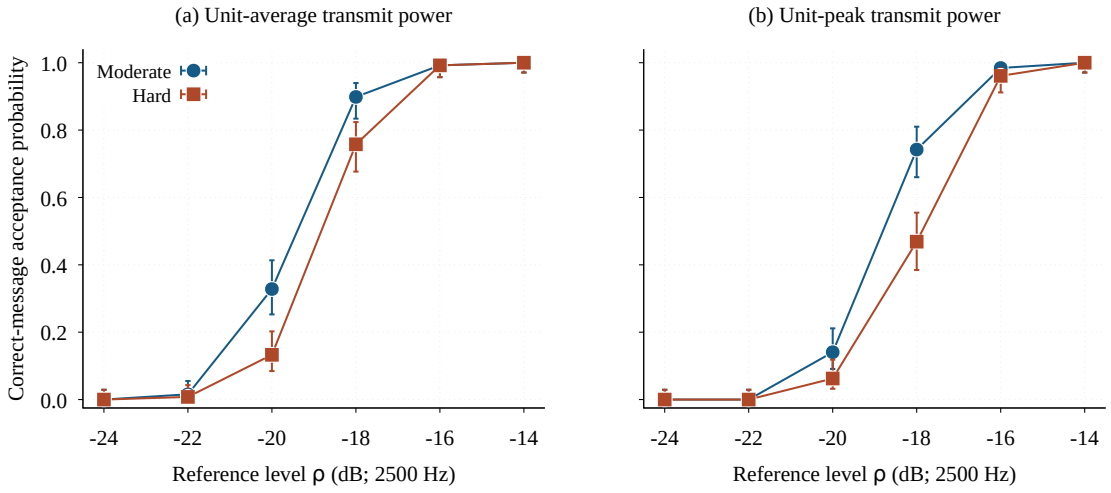


Figure 3: Correct-message acceptance with pointwise 95% Wilson intervals; 128 frames per point. Lines connect the measured points. The horizontal axis is the unit-reference level ρ. In (a) it equals average transmit SNR in 2500 Hz; in (b) average transmit SNR is approximately 0.711 dB lower. The two panels reuse the same randomized cases.

Reference ρ	Moderate: $k/128$, rate [95% CI]	Hard: $k/128$, rate [95% CI]
Unit-average power		
−24 dB	0/128, 0.000 [0.000, 0.029]	0/128, 0.000 [0.000, 0.029]
−22 dB	2/128, 0.016 [0.004, 0.055]	1/128, 0.008 [0.001, 0.043]
−20 dB	42/128, 0.328 [0.253, 0.413]	17/128, 0.133 [0.085, 0.202]
−18 dB	115/128, 0.898 [0.834, 0.940]	97/128, 0.758 [0.677, 0.824]
−16 dB	127/128, 0.992 [0.957, 0.999]	127/128, 0.992 [0.957, 0.999]
−14 dB	128/128, 1.000 [0.971, 1.000]	128/128, 1.000 [0.971, 1.000]
Unit-peak power		
−24 dB	0/128, 0.000 [0.000, 0.029]	0/128, 0.000 [0.000, 0.029]
−22 dB	0/128, 0.000 [0.000, 0.029]	0/128, 0.000 [0.000, 0.029]
−20 dB	18/128, 0.141 [0.091, 0.211]	8/128, 0.062 [0.032, 0.118]
−18 dB	95/128, 0.742 [0.660, 0.810]	60/128, 0.469 [0.384, 0.555]
−16 dB	126/128, 0.984 [0.945, 0.996]	123/128, 0.961 [0.912, 0.983]
−14 dB	128/128, 1.000 [0.971, 1.000]	128/128, 1.000 [0.971, 1.000]

Table 3: Correct acceptances and pointwise 95% Wilson intervals. Every cell has zero incorrect acceptances; its erasure count is $128 - k$. Rates are dimensionless. Full erasure and incorrect-acceptance intervals are supplied in `study/cells.csv`.

At −24 dB, none of the four conditions accepts a correct message. At −14 dB, every condition accepts all 128 messages; the Wilson interval for each 128/128 cell is approximately $[0.971, 1]$, so the observation does not establish error-free operation. The transition between these limits is resolved only at 2 dB increments. No incorrect accepted message occurs in any of the 3072 signal records; all remaining outcomes are erasures. Because the power settings are paired, those records do not constitute 3072 independent opportunities for an undetected error.

6.2 The 50% acceptance crossing

Profile	Power setting	50% crossing	95% grid bracket
Moderate	Average	−19.4 dB	$[-20, -18]$ dB
Hard	Average	−18.8 dB	$[-20, -18]$ dB
Moderate	Peak	−18.8 dB	$[-20, -18]$ dB
Hard	Peak	−17.9 dB	$[-20, -16]$ dB

Table 4: Interpolated 50% crossing in reference-level dB and conservative 95% grid brackets under a monotone-response assumption. The bracket is constructed from six simultaneous exact binomial intervals per curve. Its width reflects sampling uncertainty and grid resolution.

Table 4 places the unit-average crossings near −19.4 and −18.8 dB. Both lie between the measured −20 and −18 dB points. The moderate unit-peak crossing is also in that interval. For hard fading at unit-peak power, 60/128 messages succeed at −18 dB, giving an interpolated crossing near −17.9 dB between −18 and −16 dB. Its conservative uncertainty bracket spans $[-20, -16]$ dB because the −18 dB cell does not locate the true success probability securely on either side of 0.5. Sub-decibel interpolated values describe the chosen interpolation and should not be read as sub-decibel measurement resolution.

Lot	Captures	Raw FEC accepts	Final accepts	Gate exceedances
Calibration	2048	0	0	21
Independent check	4096	0	0	44

Table 5: Noise-only outcomes. Gate exceedance alone is not acceptance; the receiver also requires raw decoder success.

6.3 Noise-only acceptance

The independently seeded check has zero false accepts in 4096 captures, giving the upper bound in (26), or about 0.0731% per 15-second capture. The reference scales have counts 683, 683, 683, 683, 682, and 682 in increasing ρ order. The pooled bound does not supply a separate 95% upper limit of the same size at each scale. Both lots also have zero raw FEC acceptances, so the evidence-only exceedances in Table 5 must not be reported as false decoded messages.

6.4 Waveform duration, bandwidth, and peak power

The active waveform has exactly 116,800 samples at 12 kHz, including both filter tails, for a duration of 9.733333333 s. The physical summary uses 1536 unique confirmation waveforms once each; duplicated records from the two power settings do not enlarge the waveform sample. Mean B_{99} is 1958.731 Hz and the largest observed value is 1998.724 Hz. Mean PAPR is 0.710905 dB and the largest observed value is 0.710951 dB. These extrema belong to the sampled payloads, not to an exhaustive search of the codebook. The nearly fixed envelope peak makes unit-peak active-frame average power approximately 0.8490 and energy approximately 8.2636 in normalized power-seconds, compared with 1 and 9.7333 under unit-average normalization.

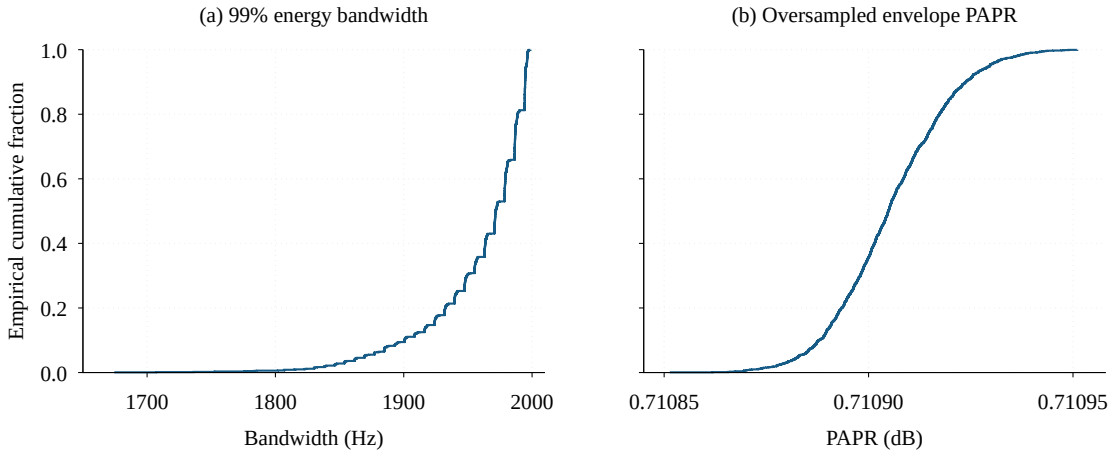


Figure 4: Empirical cumulative distributions of the measured waveform bandwidth and PAPR over 1536 unique confirmation cases. No channel fading or noise enters these transmit-waveform measurements.

7 Discussion

The observed transition from mostly erased frames near −22 dB to nearly complete acceptance at −16 dB under unit-average power is specific to the sampled channel ensemble. The hard profile combines a larger path separation and faster fading; the experiment changes both together and cannot isolate their individual effects. At −18 dB, the correct-acceptance counts are 115/128 for moderate fading and 97/128 for hard fading. The corresponding unit-peak counts are 95/128 and 60/128. Peak normalization reduces average transmit power by approximately 0.711 dB, so those two regimes must be interpreted with their distinct reference-power conventions.

The receiver integrates each symbol without a channel estimate. Doppler variation and delay-dependent spectral cancellation can redistribute energy within and between its tone projections. The nonbinary decoder combines these soft observations, but the fixed metric parameter γ and the fixed pilot threshold are not adapted to the individual fading realization. The present data characterize their joint behavior; they do not identify which stage caused each erased frame.

The guard-band estimator relies on noise in the spectral guards representing noise in the tone region. The study uses full-band complex Gaussian noise. Colored interference, impulsive noise, carrier drift, and other delay/Doppler profiles were not sampled. The WAV frontend reconstructs an analytic signal and removes part of the complex-IQ spectral support, so the reported IQ noise statistics do not automatically describe an audio channel. These are specific changes to the observation model that require separate measurements.

A fixed-seed estimator validation used 128 cases at every point in Table 6. The AWGN rows use the known transmitted payload, so they measure the estimator without conditioning on decoder success. The moderate and hard rows use only CRC-valid correct receiver outputs; the valid count therefore records the available estimates. Bias is the mean of $\widehat{\mathrm{SNR}}_{2500}$ minus the configured SNR. The AWGN absolute bias remains below 0.08 dB. Delay and Doppler spread energy outside the nominal correct-tone projections, causing the increasing negative bias in the HF profiles. The implementation applies no empirical offset to remove it.

Channel	Target (dB)	Valid / 128	Bias (dB)	RMSE (dB)
AWGN	−24	128	−0.073	0.892
AWGN	−22	128	−0.080	0.627
AWGN	−20	128	−0.059	0.464
AWGN	−18	128	−0.023	0.439
AWGN	−16	128	−0.059	0.286
AWGN	−14	128	−0.026	0.234
Moderate	−18	115	−0.166	0.925
Moderate	−16	127	−0.250	1.029
Moderate	−14	128	−0.226	1.002
Hard	−18	104	−0.764	1.049
Hard	−16	128	−0.963	1.318
Hard	−14	128	−1.009	1.270

Table 6: Fixed-seed validation of the 2500 Hz SNR estimate.

No incorrect accepted messages occurred in the signal records, and no raw or gated decoder acceptances occurred in the independent noise-only set. Thus these records bound observed acceptance errors under the stated design, but do not demonstrate an incremental reduction caused by the pilot gate. The noise bound is pooled over six reference noise scales and is not a uniform per-scale bound. The confirmation payload ensemble also fixes six bits;

tests of arbitrary 77-bit inputs establish functionality, while error rates over the full payload space remain unmeasured.

8 Implementation and reproducibility

The standalone distribution supplies Fortran source, command-line programs, and static and shared libraries. Headerless float32 little-endian IQ stores interleaved I/Q samples at 12 kHz. The optional PCM16 mono WAV frontend uses a selectable audio center, 1500 Hz by default. The interface guide defines file bounds, result fields, the C layout, and serialized access to the shared FFT cache [7]. These interface details do not enter the statistical denominators.

The study directory contains the per-trial signal and noise records, their checksums and provenance, the channel implementation and single-trial reproduction harness, and a Fortran analysis program. Its commands regenerate the cell tables and plotting data; the plotting script renders those results without recomputing the statistics [8].

A clean GNU Fortran/C build and the regression suite verify the implementation separately from the performance study. The suite passes 88 core checks, 26 I/O checks, and 39 CLI checks, together with Fortran and C API roundtrips. These are assertion counts, not channel trials. The fixed-seed SNR validation is a separate executable and reports all cases in Table 6. The regression suite also matches the retained codeword, complete CF32 frame, and PCM waveform fixtures. The exact CF32 roundtrip checks the integer representations of quantized samples. Strict compilation treats compiler warnings as errors [9].

The code is distributed under GPL-3.0-or-later. Nico Palermo's graph, coefficient, algorithm and metric attribution and the relevant source-package identifiers are retained in `licenses/QRA-NOTICE.txt`. Checksums identify the delivered source and data.

9 Conclusion

QF256 transmits a 77-bit message in a 9.733333-second filtered 256-tone burst and combines blind pilot acquisition with nonbinary soft decoding and a fixed acceptance rule. CRC-valid outputs include a 2500 Hz-referenced SNR estimate whose AWGN behavior and dispersive-channel bias are quantified above. The completed HF study provides cell-level success estimates and uncertainty under two specified fading profiles and two power constraints. The independent noise check provides a finite-sample bound on false acceptance, while waveform measurements quantify bandwidth and peak power. The accompanying records and analysis code reproduce the reported tables and figures.

A Graph specification

The input indices v_i, $i = 1, \ldots, 51$, for (2)–(3) are the following sequence, read left to right and top to bottom:

```
14, 2, 4, 5, 9, 13, 10, 15, 11, 6, 1, 8, 2, 12, 9, 10, 13,
7, 4, 11, 8, 6, 3, 14, 13, 5, 9, 1, 2, 12, 3, 10, 15, 6,
7, 14, 8, 13, 12, 3, 10, 1, 11, 5, 8, 15, 9, 12, 4, 7, 11
```

The coefficient exponents ℓ_i are:

```
0, 14, 0, 0, 13, 37, 0, 27, 56, 62, 29, 0, 52, 34, 62, 4, 3,
22, 25, 0, 22, 0, 20, 10, 0, 43, 53, 60, 0, 0, 0, 62, 0, 5,
0, 61, 36, 31, 61, 59, 10, 0, 29, 39, 25, 18, 0, 14, 11, 50, 17
```

To construct h_i, first generate the powers of element 2 in GF(64) using polynomial `0x43`, and take the integer value represented by the six-bit element at exponent ℓ_i. Embed that integer as the corresponding eight-bit polynomial element of GF(256). The numbers ℓ_i are therefore *not* used directly as logarithms in GF(256). The literal arrays and this conversion are preserved in the shipped coding source [6].

References

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[6] QF256, *Nonbinary coding implementation*, version 1.0.0, `src/core/qra_soft.f90`.

[7] QF256, *Standalone implementation and interface documentation*, version 1.0.0, `README.md`, `docs/API.md`, `src/qf256_api.f90`, and `include/qf256.h`.

[8] QF256, *HF confirmation and noise-only study records and analysis*, October 2026, `study/`. Data, source and reproduction instructions in the accompanying distribution.

[9] QF256, *Build, regression and source-integrity records*, October 2026, `docs/VALIDATION.txt`, `docs/CORE.sha256`, and `docs/DELIVERY.sha256`.

[10] N. Palermo, `qra15_65_64_irr_e23.c` (2020) and `qracodes.c` (2016), QRA graph and decoder sources, GPL-3.0-or-later. Source package: WSJT-X 2.7.0+repack-1, Debian Sources, lib/qra/q65, retrieved October 10, 2026. Attribution and encoding references are preserved in `licenses/QRA-NOTICE.txt`.